



Damascus University
Faculty of Informatics
Formal Languages and Automata Course



Introduction and Overview

Lecture 1

What are the fundamental capabilities and limitations of computers?

Theory of computing

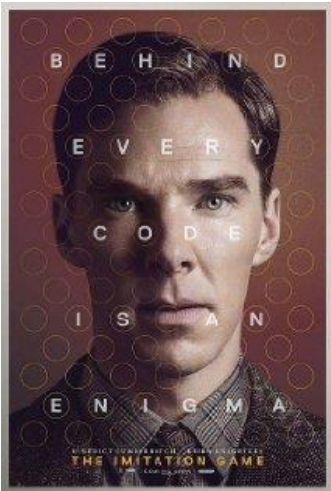
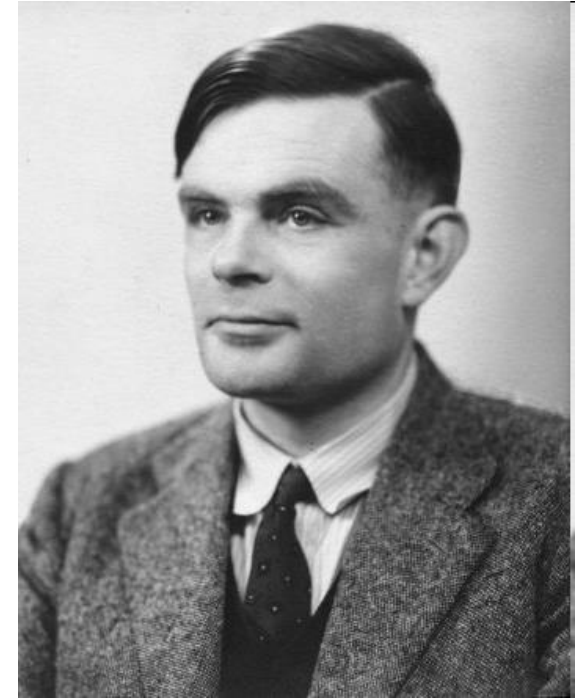
COMPLEXITY
THEORY

computability
theory

AUTOMATA
Theory

(A pioneer of automata theory) Alan Turing (1912-1954)

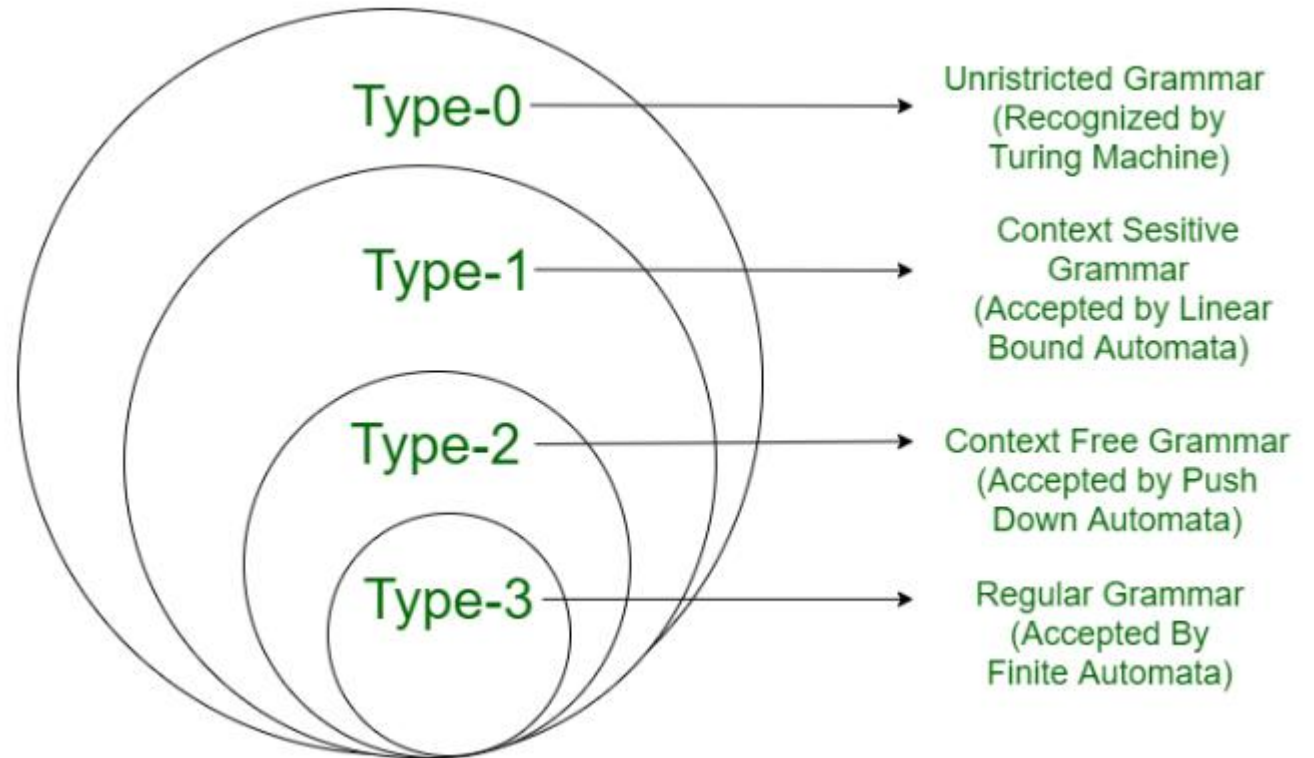
- Father of Modern Computer Science
- English mathematician
- Studied abstract machines called *Turing machines* even before computers existed



AUTOMATA THEORY

- Automata theory deals with the definitions and properties of mathematical models of computation.
- These models play a role in several applied areas of computer science.
- One model, called the *finite automaton*, is used in text processing, compilers, and hardware design.
- Another model, called the *context-free grammar*, is used in programming languages and artificial intelligence.

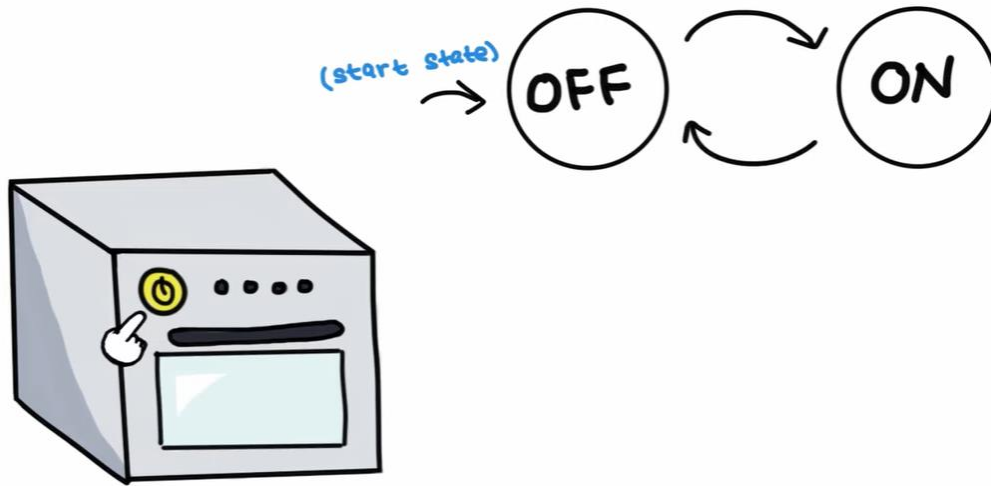
The Chomsky hierarchy



The Chomsky hierarchy

Type	Grammar	Automaton	Language Class	Use cases
0	Unrestricted	Turing Machine (unlimited memory)	Recursively Enumerable	Theoretical computation (e.g., solving the Halting Problem).
1	Context- Sensitive	Linear-Bounded Automaton (memory proportional to input length)	Context-Sensitive	Natural language syntax, programming language features
2	Context-Free	Pushdown Automaton (stack memory)	Context-Free	Programming language grammars (e.g., arithmetic expressions)
3	Regular	Finite-State Automaton (no memory)	Regular	Pattern matching (e.g., regex for text processing)

FSM – Finite State Machine



- Simplest computational model that we can construct
- It has limited memory
- Very often used in every day life

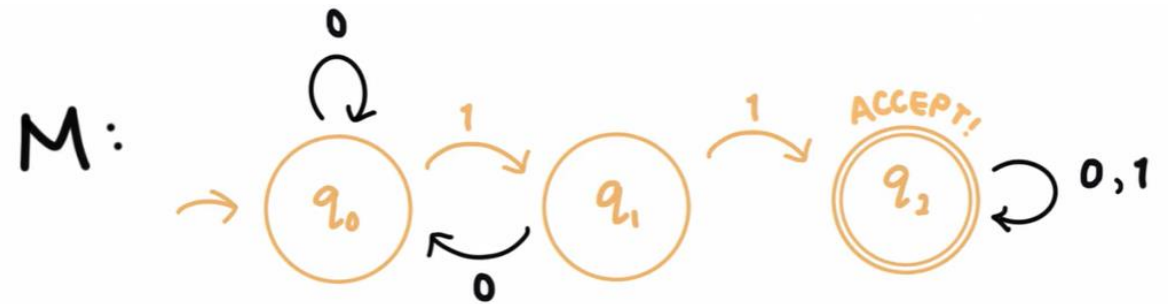
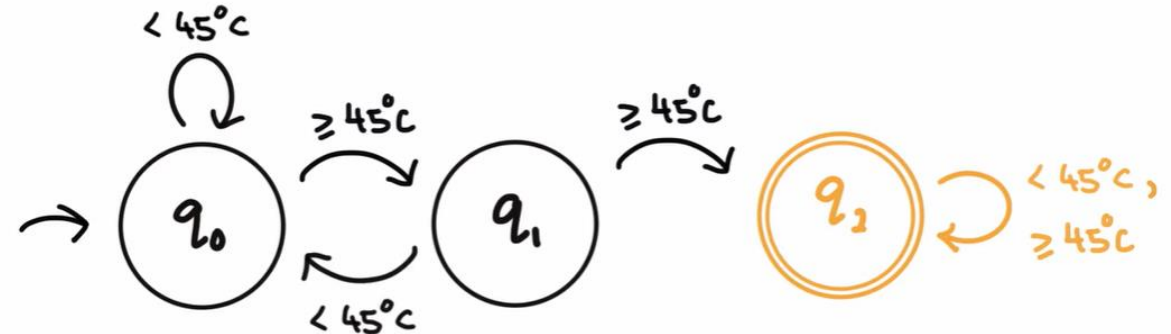
FSM – Finite State Machine

- **Problem: Design a Finite State Automaton (FSA) to Detect Heatwaves**

Definition: A heatwave occurs when temperatures reach $\geq 45^{\circ}\text{C}$ (113°F) for **two consecutive days**.

- **States:**

- **q0:** Initial state (no consecutive $\geq 45^{\circ}\text{C}$ days).
- **q1:** One day with $\geq 45^{\circ}\text{C}$.
- **q2:** Accepting state (heatwave detected: two consecutive $\geq 45^{\circ}\text{C}$ days).

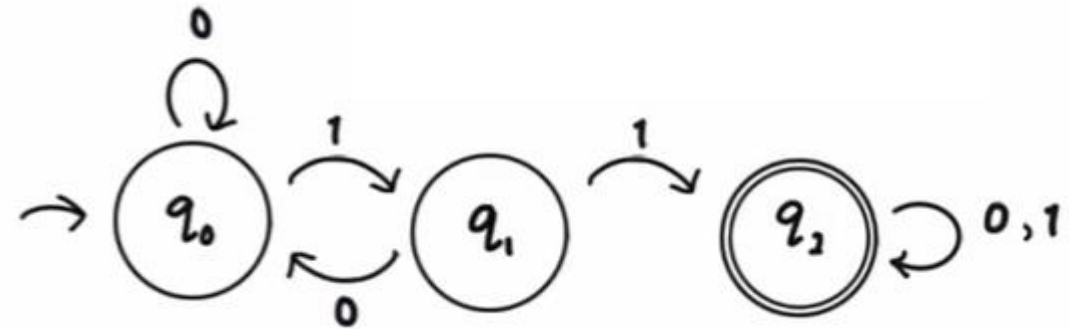


FSM – Finite State Machine

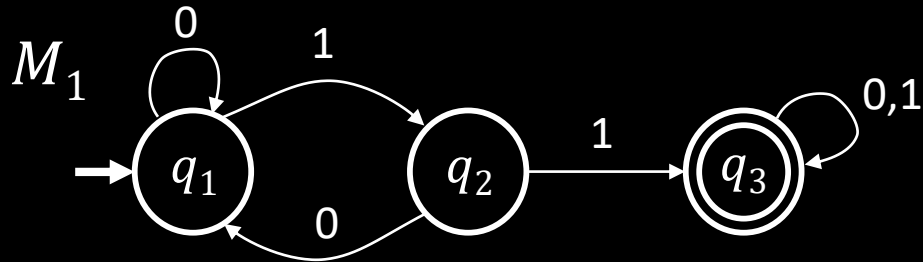
This automata accepts all binary strings that contains 11
 $L(M) = \{\text{All binary strings containing the substring "11"}\}$

- **Acceptance Examples:**

- M Accept 11
- M Accept 110
- M Accept 0110



Finite Automata



States: $q_1 q_2 q_3$

Transitions: $\xrightarrow{1}$

Start state: $\rightarrow \bigcirc$

Accept states: $\bigcirc\bigcirc$

Input: finite string

Output: Accept or Reject

Computation process: Begin at start state, read input symbols, follow corresponding transitions, Accept if end with accept state, Reject if not.

Examples: 01101 $\rightarrow$ Accept

00101 $\rightarrow$ Reject

M_1 accepts exactly those strings in A where
 $A = \{w \mid w \text{ contains substring } 11\}.$

Say that A is the language of M_1 and that M_1 recognizes A and that $A = L(M_1).$

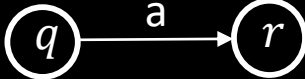
Finite Automata – Formal Definition

Defn: A finite automaton M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$

Q finite set of states

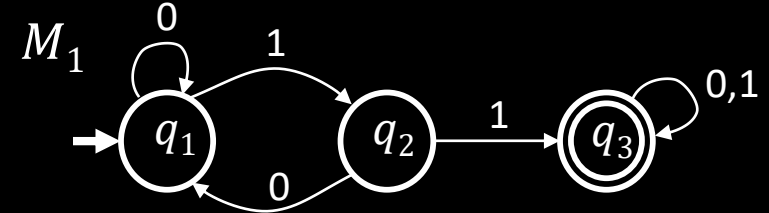
Σ finite set of alphabet symbols

δ transition function $\delta: Q \times \Sigma \rightarrow Q$

q_0 start state $\delta(q, a) = r$ means 

F set of accept states

Example:



$$M_1 = (Q, \Sigma, \delta, q_1, F)$$

$$Q = \{q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$

$$F = \{q_3\}$$

$\delta =$	0	1
q_1	q_1	q_2
q_2	q_1	q_3
q_3	q_3	q_3

Finite Automata – Computation

Strings and languages

- A string is a finite sequence of symbols in Σ
- A language is a set of strings (finite or infinite)
- The empty string ε is the string of length 0
- The empty language $\emptyset$ is the set with no strings

Defn: M accepts string $w = w_1w_2 \dots w_n$ each $w_i \in \Sigma$ if there is a sequence of states $r_0, r_1, r_2, \dots, r_n \in Q$ where:

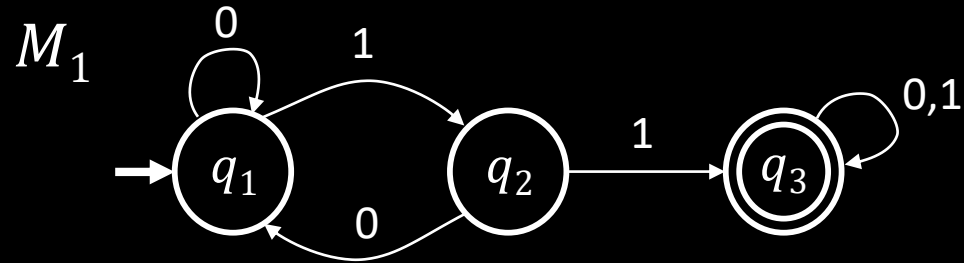
- $r_0 = q_0$
- $r_i = \delta(r_{i-1}, w_i)$ for $1 \leq i \leq n$
- $r_n \in F$

Recognizing languages

- $L(M) = \{w \mid M \text{ accepts } w\}$
- $L(M)$ is the language of M
- M recognizes $L(M)$

Defn: A language is regular if some finite automaton recognizes it.

Regular Languages – Examples



$L(M_1) = \{w \mid w \text{ contains substring } 11\} = A$

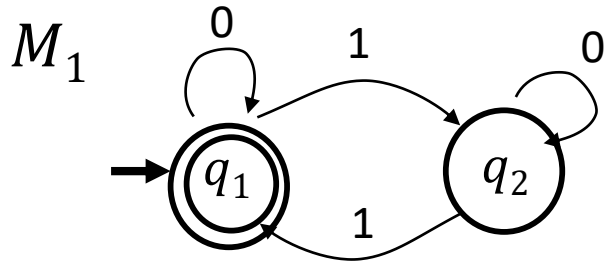
Therefore A is regular

More example:

Let $B = \{w \mid w \text{ has an even number of 1s}\}$
 B is regular (make automaton for practice).

Regular Languages – Example1

Let $B = \{w \mid w \text{ has an even number of 1s}\}$
 B is regular (make automaton for practice).



References

- MIT OpenCourseWare <https://ocw.mit.edu>
- Michel Spiser , Introduction to theory of computation , 3rd edition